BFG:XR Alternate Activation: Draft Options

Companion to the BFG:XR condensed booklet. Kept separate from the booklet because the alternate activation (AA) rules are not decided. Drafted 12 November 2026; updated 13 November 2026 with the ABFG text and Warp Rift articles.

Status: draft for discussion, not rules. This document sets out three ways of deciding activation order under AA, with the numbers we worked out for each. The common framework in section 1 is taken from the AA working notes; items marked OPEN there are still open. Section 2 summarises ABFG v0.3; rule wording in sections 3-4 is proposed wording, not settled text.

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1. Common framework (from the AA working notes)

These points are shared by all three systems. They come from the AA working notes; none of the three ordering systems changes them.

- **Scope.** The base game stays IGOUGO. AA lives in its own self-contained section that redefines the turn sequence only.
- **Phased turn, shared phases.** Both sides act inside each shared phase. Each ship or squadron activates once per phase. All movement resolves before all shooting, and so on.
- **Working round structure:** 0 Command phase (special orders declared); 1 Ordnance phase A (on-board ordnance moves); 2 Shooting phase; 3 Movement phase; 4 Ordnance phase B (on-board ordnance moves); 5 End phase. OPEN Whether shooting comes before or after movement is still open.
- **Special orders.** Declared in the command phase; they last until the ship's next activation. Brace for Impact is a pre-committed command-phase order, with its full penalty for the whole turn.
- **Ordnance.** All ordnance gets 1 move on its first turn on the board and 2 moves every turn after that.
- **Eldar** use MMS v2.0 under AA.
- **Boarding and ramming.** The lock applies when the board or ram is declared in the movement phase; the combat resolves in the end phase in initiative order. A ship boarded or rammed before it has activated loses its move.
- OPEN End-phase cadence (blast marker removal, critical repair and fire damage happening half as often per round), blast markers created mid shooting phase, ordnance-phase sequencing, and a global "turn means round" ruling are still open in the working notes.

Activation unit. In all the numbers below, each capital ship and each escort squadron is one unit with one activation per phase. Capital ships in a squadron are counted separately here. OPEN The working notes do not yet define the activatable unit formally.

2. System 1: fixed initiative order (ABFG)

From *A Big Fleet Game* (ABFG) v0.3, a fan AA conversion of BFG. This section summarises ABFG's own text. The initiative formula did not survive in our text copy of ABFG (it was an image); the formula below is the one recorded in our AA working notes from ABFG.

> **Initiative.** Each unit's initiative = Leadership − starting hits + 10 (higher is better), set at the start of the game. A capital squadron uses the highest Leadership and the highest hits in the squadron, which may come from different ships. An escort squadron totals its ships' hits.
>
> **Changes.** Leadership changes during the game (for example Bridge Smashed, or area effects) change initiative. A crippled ship loses 1 initiative. Changes take effect only at the start of the next game turn.
>
> **Order.** Shooting phase: units activate in initiative order (highest first). Movement phase: **inverse** initiative order (lowest first), so high-initiative units move last, after seeing the enemy commit.

Ties. Broken by Leadership, then the admiral's Leadership, then fleet strategy rating, then a dice-off each turn.

Forcing an activation. At the start of the shooting or movement phase a player may spend a re-roll to make one ship, friendly or enemy, activate first. If both players do, a dice-off (2D6 + the highest Leadership in the fleet, which admiral re-rolls can influence) decides which goes first. ABFG suggests doubling re-rolls as a test, since they get more use.

Turn. Command phase (face-down order tokens, including Carry On and Disengage), shooting, movement, ordnance, end. Orders are tested when the unit activates; a failure becomes Carry On and does not stop other ships from trying. Damage, criticals, Brace saves and holofield saves are all applied in the end phase, in initiative order. In the ordnance phase a dice-off decides who moves first, and ordnance moves 1-2-2-1.

Our framework (section 1) differs from ABFG in some places: movement before or after shooting is open, and damage is applied immediately. System 1 here means ABFG's ordering rules on our framework.

Pros

- Every ship has its own place in the order, so Leadership and ship size matter ship by ship.
- The two orders are well matched to the phases: the better unit shoots first and moves last.
- No randomness in the order: players can plan around it.

Cons

- Accounting: every unit's initiative is worked out once and the units ranked, then rechecked at the start of each turn for crippling and Leadership changes, with a tie-break chain. Lighter than I first described, but still the heaviest of the three. You rejected ABFG earlier as "too much accounting".
- Predictable, and therefore plannable by the opponent.
- A strong edge: with our test fleets the leading side gets an edge share of 0.66-0.99 when it leads by 3 points or more. Near 1, every unit of one fleet shoots before every unit of the other, and moves after it.
- ABFG can afford this because damage waits for the end phase. With damage resolving immediately, an edge this strong becomes an alpha strike.

3. System 2: vanilla dice in a bag

Bag. At the start of each phase, each side puts one die of its colour in the bag for each unit it has that can activate in that phase.

Draw. Draw one die at a time. The side whose colour is drawn activates one of its units that has not yet acted this phase. Repeat until the bag is empty, then move to the next phase.

Pros

- No calculation at all, and nothing to rank or track.
- Cannot be gamed the way strict alternation can.
- Every unit stays a live option until it acts.

Cons

- No fleet gets any edge for better crews or smaller, more agile ships: the edge share is 0 for every matchup.
- The side with more units tends to have units left at the end of a phase, which lets it act after the opponent has committed. That is a real edge in movement, but it comes from unit count rather than quality.

4. System 3: dice in a bag with initiative adjustment

System 2's bag, with a fleet-level initiative edge that grows with the difference between the two fleets' mean initiative.

Unit initiative. As in System 1: Leadership − starting hits + 10, with escort squadrons using their single rolled Leadership and summed starting hits. A ship carrying a Necron Obelisk uses the Leadership it rolls, as normal.

Fleet mean. Each fleet's mean initiative is the average over its units, worked to one decimal place. The **gap** is the difference between the two means. The fleet with the higher mean is the **edge holder**.

When it is worked out. At the start of the game, after Leadership is rolled, using starting hits. OPEN Options: recompute only when a unit is destroyed; or follow ABFG and apply Leadership changes and a −1 for each crippled ship at the start of each game turn. Never use current hits (lost hits would raise initiative).

Gap	Result	Edge share (test fleets)
Under 0.5	No edge holder: use the bag as normal (System 2).	0
0.5 to under 1.5	The edge holder chooses which side takes the first activation in each phase.	0.07-0.50
1.5 to under 3	The choice, plus 1 extra die per 3 units.	0.33-0.67
3 or more	The choice, plus 1 extra die per 2 units.	0.40-0.60

Counting. Extra dice are counted from the edge holder's own fleet, rounding up. An escort squadron counts as one unit.

The choice. At the start of each phase the edge holder decides which side takes the first activation. That side activates one unit without a draw, and **removes one of its own dice from the bag**. Every activation spends one of that side's dice, including the free first one.

Extra dice. Extra dice go in the bag in the edge holder's colour, and the bag is refilled at the start of each phase. They give no extra activations. Once a side has activated all its units in a phase, any further draws of its colour do nothing.

Passing. PROPOSED A side may decline a draw of its colour (pass) while it still has more dice in the bag than units left to activate. This is how the extra dice let the edge holder hold back in movement as well as act early in shooting.

Why a fixed bonus per number of units

A flat bonus (such as 1 extra die, or simply going first) is worth less the bigger the fleets are. One extra die among 34 barely registers. The edge from extra dice depends on the ratio of extra dice to units, so giving 1 die per N units keeps the edge about the same at any game size:

Units (Eldar v Imperial)	Flat +1 die	Go first only	+1 per 2 units	+1 per unit
3 v 3	0.20	0.33	0.33	0.43
5 v 6	0.14	0.20	0.33	0.45
10 v 12	0.08	0.10	0.31	0.48
15 v 18	0.06	0.07	0.33	0.48

Choosing who goes first is worth about 1 divided by the edge holder's number of units, so it is strongest in small games (0.33 at 3 units, 0.10 at 10). That is why the table gives it alone only for small gaps, and why the dice rates are one step lower than they would be without it.

Why the edge is aimed below ABFG's

We measured activation order only. ABFG can afford a large ordering edge because its damage resolves in the end phase. In a bag system where shots take effect immediately, the same ordering edge is worth more in results, so the table aims for a moderate edge (roughly 0.3-0.6) rather than ABFG's 0.66-0.99 at gaps of 3 or more. The right size can only be found by playing.

Results on the test fleets

Points	Imperial list (units)	Eldar list (units)	Gap	ABFG	System 3 gives	System 3
500	gunline (3)	gunline (3)	+3.0	+0.78	Eldar: choice + 2	+0.60
500	gunline (3)	escort (3)	+4.7	+0.93	Eldar: choice + 2	+0.60
500	carrier/escort (4)	gunline (3)	+1.0	+0.32	Eldar: choice	+0.33
500	carrier/escort (4)	escort (3)	+2.7	+0.70	Eldar: choice + 1	+0.50
1000	original (6)	original (5)	+3.0	+0.78	Eldar: choice + 3	+0.50
1500	gunline (7)	gunline (6)	+3.2	+0.66	Eldar: choice + 3	+0.44
1500	gunline (7)	escort (8)	+4.7	+0.84	Eldar: choice + 4	+0.42
1500	carrier/escort (8)	gunline (6)	+2.1	+0.51	Eldar: choice + 2	+0.38
1500	carrier/escort (8)	escort (8)	+3.6	+0.78	Eldar: choice + 4	+0.42
2000	gunline (8)	gunline (7)	+3.3	+0.71	Eldar: choice + 4	+0.45
2000	gunline (8)	escort (10)	+5.7	+0.99	Eldar: choice + 5	+0.40
2000	carrier/escort (10)	gunline (7)	+1.3	+0.34	Eldar: choice	+0.14
2000	carrier/escort (10)	escort (10)	+3.7	+0.87	Eldar: choice + 5	+0.40

Gap = Eldar mean initiative − Imperial mean initiative, each to one decimal. Positive edge shares favour the Eldar. Shooting phase; see Appendix A. All 91 matchups, including the escort builds below, are in Appendix C.

Escort squadron size as a build strategy

Because a squadron's initiative uses its summed hits, how a player squadrons the same escorts changes the fleet's mean initiative. To test this, each faction has an escort-heavy list at each points level built three ways from the same ships: massed into squadrons of six, in squadrons of three or four, or in pairs. Your estimate (no data) is that pairs, threes and fours are about equally common in vanilla play; Orks tend to blob. Pairs usually come from cruiser-heavy fleets spending leftover points that won't buy a cruiser, more than from escort fleets split up. In that case one pair raises the fleet's mean initiative by 0.65-0.9 (Imperial 1500 gunline +0.9, Eldar 1500 and 2000 gunlines +0.65 and +0.75), close to a full band, for 60-150 points. That is the effect the working notes intended: it gives cruiser-heavy fleets a reason to bring escorts. One class per squadron. The lists are in Appendix B.

Mean initiative, same ships:

Points	Imperial: massed (6s)	Imperial: 3s and 4s	Imperial: pairs	Eldar: massed (6s)	Eldar: 3s and 4s	Eldar: pairs
500	11.5 (3 units)	13.9 (5 units)	14.9 (7 units)	13.5 (2 units)	15.2 (3 units)	16.0 (4 units)
1000	11.4 (5 units)	13.7 (8 units)	14.7 (11 units)	13.6 (4 units)	14.6 (5 units)	15.7 (7 units)
1500	10.6 (7 units)	12.1 (9 units)	13.1 (11 units)	13.6 (6 units)	14.3 (7 units)	15.6 (10 units)
2000	10.1 (9 units)	12.0 (12 units)	13.1 (15 units)	12.9 (7 units)	14.6 (10 units)	15.5 (13 units)

Going from massed to pairs raises a fleet's mean initiative by 2.0-3.4; squadrons of three or four sit in between, 0.7-2.4 above massed. Splitting works in two ways: a pair's initiative is high (Ld + 8), and the extra units dilute the low-initiative capital ships in the average. That swing is as large as the whole Eldar-Imperial difference between lists built the same way (a gap of 1.0-3.0).

Each escort build against each other:

Points	Imperial escorts	Eldar escorts	Gap	ABFG	System 3 gives	System 3
500	pairs	pairs	+1.1	+0.55	Eldar: choice	+0.25
500	pairs	massed (6s)	-1.4	-0.40	Imperial: choice	-0.14
500	pairs	3s and 4s	+0.3	+0.22	No edge holder: plain bag	0.00
500	massed (6s)	pairs	+4.5	+0.99	Eldar: choice + 2	+0.50
500	massed (6s)	massed (6s)	+2.0	+0.83	Eldar: choice + 1	+0.67
500	massed (6s)	3s and 4s	+3.7	+0.98	Eldar: choice + 2	+0.60
500	3s and 4s	pairs	+2.1	+0.69	Eldar: choice + 2	+0.50
500	3s and 4s	massed (6s)	-0.4	-0.01	No edge holder: plain bag	0.00
500	3s and 4s	3s and 4s	+1.3	+0.66	Eldar: choice	+0.33
1000	pairs	pairs	+1.0	+0.45	Eldar: choice	+0.14
1000	pairs	massed (6s)	-1.1	-0.35	Imperial: choice	-0.09
1000	pairs	3s and 4s	-0.1	+0.02	No edge holder: plain bag	0.00
1000	massed (6s)	pairs	+4.3	+0.99	Eldar: choice + 4	+0.45
1000	massed (6s)	massed (6s)	+2.2	+0.91	Eldar: choice + 2	+0.50
1000	massed (6s)	3s and 4s	+3.2	+0.98	Eldar: choice + 3	+0.50
1000	3s and 4s	pairs	+2.0	+0.59	Eldar: choice + 3	+0.40
1000	3s and 4s	massed (6s)	-0.1	+0.07	No edge holder: plain bag	0.00
1000	3s and 4s	3s and 4s	+0.9	+0.45	Eldar: choice	+0.20
1500	pairs	pairs	+2.5	+0.54	Eldar: choice + 4	+0.36
1500	pairs	massed (6s)	+0.5	-0.04	Eldar: choice	+0.17
1500	pairs	3s and 4s	+1.2	+0.16	Eldar: choice	+0.14
1500	massed (6s)	pairs	+5.0	+0.89	Eldar: choice + 5	+0.40
1500	massed (6s)	massed (6s)	+3.0	+0.73	Eldar: choice + 3	+0.44
1500	massed (6s)	3s and 4s	+3.7	+0.79	Eldar: choice + 4	+0.45
1500	3s and 4s	pairs	+3.5	+0.65	Eldar: choice + 5	+0.40
1500	3s and 4s	massed (6s)	+1.5	+0.27	Eldar: choice + 2	+0.38
1500	3s and 4s	3s and 4s	+2.2	+0.46	Eldar: choice + 3	+0.40
2000	pairs	pairs	+2.4	+0.55	Eldar: choice + 5	+0.33
2000	pairs	massed (6s)	-0.2	-0.14	No edge holder: plain bag	0.00
2000	pairs	3s and 4s	+1.5	+0.30	Eldar: choice + 4	+0.36
2000	massed (6s)	pairs	+5.4	+0.98	Eldar: choice + 7	+0.40
2000	massed (6s)	massed (6s)	+2.8	+0.89	Eldar: choice + 3	+0.40
2000	massed (6s)	3s and 4s	+4.5	+0.97	Eldar: choice + 5	+0.40
2000	3s and 4s	pairs	+3.5	+0.67	Eldar: choice + 7	+0.40
2000	3s and 4s	massed (6s)	+0.9	+0.18	Eldar: choice	+0.14
2000	3s and 4s	3s and 4s	+2.6	+0.62	Eldar: choice + 4	+0.36

- Across the common builds (pairs, 3s and 4s, on either side), the gap runs from −0.1 to 3.5 and System 3 gives 0 to 0.50. The Eldar never lose the edge, but Imperial pairs against Eldar 3s and 4s remove it at 500 and 1000 points.
- Squadron size can decide who holds the edge more than faction does. Imperial pairs against massed Eldar take the edge from the Eldar at 500 and 1000 points, cut it to the bare choice at 1500, and remove it at 2000.
- When both sides split, the Eldar keep a smaller edge (gap 1.0-2.5).
- System 3 damps the effect: the swing between the best and worst build is −0.14 to +0.67, against −0.40 to +0.99 under ABFG. Extra units also earn more dice, but the dice rate is per unit, so the edge share hardly changes with unit count; the effect works through the gap.

You have treated splitting as intended: dispersed escorts cover more space. The counterweights are outside initiative. Pairs concede VP sooner, under both XR and Remastered scoring:

Escorts killed (6 Swords, 210 pts)	1	2	3	4	6
XR: one squadron of 6	0	0	52.5	52.5	210
XR: three pairs (opponent targets well)	17.5	70	87.5	140	210
Remastered: one squadron of 6, not disengaged	0	0	0	0	210
Remastered: three pairs, not disengaged	0	70	70	140	210

A pair also loses half its firepower and its turret support from a single kill. In its favour, it has more activations, which lets a side hold units back and act after the opponent. Whether pairs are simply better is untested (section 7). If they are, every player will split every squadron and the choice disappears. The fallbacks are mean Leadership only, or averaging over individual ships so that squadron size no longer matters.

5. Comparing the three systems

	System 1: fixed order	System 2: vanilla bag	System 3: adjusted bag
Edge for the leading fleet (test fleets)	up to 0.99, growing with the gap but varying a lot with list composition	0	up to 0.67, in steps by gap band
Mental load	High: rank every unit, recheck each turn, tie-break chain	None	Low: one mean per fleet, one table lookup per game
Randomness in order	None (ties only)	Full	Full, tilted towards the edge holder
Per-ship identity	Yes: each ship's own initiative sets its slot	No	No: fleet-level edge only
Behaviour at large gaps	Near-total: one fleet shoots entirely first and moves entirely last	No change	Capped at about 0.4-0.6 (0.67 for a 2-unit fleet)
Gameability	Predictable order can be planned around	Low	Low in play; escort splitting is a list-building lever (section 4)

Edge share, used throughout: 0 means a fair bag; 1 means every unit of the edge holder acts before every unit of the opponent. See Appendix A.

6. Other ideas considered

Idea	What it was	Where it ended up
Strict alternation	Sides take turns activating one unit each	Rejected: can be gamed
Idea A (working notes)	Edge holder gets 1 extra die, whatever the gap or fleet size	Too weak and fades with fleet size (0.14 at 5 units, 0.06 at 15); superseded by System 3
Idea B (working notes)	The worse-initiative side gets the movement-phase die	Rejected in the notes: it only shoves a side late
Seize token (working notes)	One "seize" per phase instead of a die: activate now in shooting, or force an enemy unit to activate now in movement	Recorded alternative if dead draws annoy
Returned die	The extra die goes back in the bag once or twice per round	A modest increase (rank difference 1.23-1.47 against 0.78 for Idea A); superseded
Nimble tokens / four colours	Each side has a "nimble" and a "line" colour; nimble units put in 2 dice and may pass	Rejected: the extra complication isn't worth it. It sorted ships within a fleet (nimble units averaged about 2.4 ranks earlier), but the hits term in the initiative mean already rewards small ships
Go first + 1 die	Edge holder always goes first and gets 1 extra die	Equal to 3 extra dice at 5 units (0.33); fades with size
Decide the first fraction	Edge holder decides the first quarter, third or half of all activations	Rejected: a deterministic opening block is the sharpest form of alpha strike (half gives 1.00)
Long ladder	Eight 0.5-wide gap bands, up to +3 dice per unit	Tracks ABFG more closely (average error 0.10), but the precision exceeds what the model supports, and bags reach 40-50 dice
Forcing an activation (ABFG)	Spend a re-roll at the start of a phase to make one ship, friendly or enemy, activate first; dice-off if both players do	Could sit on top of any system as a re-roll use
Mean Leadership only	Use mean Ld instead of mean initiative	Simplest and can't be gamed; loses the small-ships-are-nimble effect. Fallback if escort splitting proves too strong

Simultaneous movement and combat (Warp Rift 19, 24 and 25)

Reg Steiner's *Simultaneous Movement and Combat Rules* (Warp Rift 19, pp.9-11; corrected sequence in Warp Rift 24, p.14; Eldar rule in Warp Rift 25, p.16) is not alternate activation: both players act at once each phase. It deals with the same problem, the player who shoots first winning.

- **Orders.** Both players place special orders, then a face-down "routine" counter by each ship or squadron: straight or turn. Failed special orders are replaced by a routine counter. Counters are revealed and all ships move as ordered, at least their minimum distance.
- **Reaction rule.** Seeing the enemy move, a player may test a ship's own Leadership (no admiral re-rolls) to switch its routine order (straight to turn or vice versa). Special orders can't be changed. One reaction attempt per ship per movement phase.
- **Combat** (Warp Rift 24 sequence): fire at ordnance launched in earlier turns and apply damage; fire at enemy ships with the remaining weapons and apply damage; launch new ordnance and resolve its attacks; move ordnance already in play. Both players fire together and damage is applied after each step, so a ship crippled in a step still returns fire in that step.
- **Optional initiative die** (Warp Rift 19): each ship about to fire rolls a D6 against its opponent's D6. The higher roll's damage, criticals and crippling apply before the lower roll's ship fires. The roll is never modified.
- **Eldar** (Warp Rift 25): before combat, the Eldar's opponent tests Leadership −1 for each ship. If it fails, the Eldar ship fires first and makes its second move before being shot. If it passes, both fire together. The editor's note suggests Eldar MMS as an alternative under the simultaneous rules.

Relevance here. It removes "who goes first" rather than adjusting it, at the cost of hidden orders and a reaction test. Its Eldar problem matches our reason for using MMS under AA: a shared combat step collapses the gap Move-Shoot-Move relies on. The per-ship initiative die is a possible light tie-break if two ships' activations clash.

7. Open questions and tests

- **Movement value.** Every number here covers the shooting phase. In movement, acting later (seeing the enemy commit) is often worth more than acting first. The choice in System 3 lets the edge holder use that, but the calculations can't measure it.
- **Edge size.** Whether 0.3-0.6 is the right target depends on how much acting first decides games when damage is immediate. Playtest or simulate.
- **Escort splitting.** Splitting swings mean initiative by 2-3.4 (section 4). Play the same list twice, once with escorts in pairs and once in full squadrons. If pairs win clearly, switch to mean Leadership or average over individual ships.
- **Recalculation timing.** Fixed at the start of the game with a recheck only when a unit is destroyed, or something else.
- **Passing rule** (section 4) needs confirming.
- **ABFG initiative formula.** Our text copy of ABFG lost the formula; confirm Ld − starting hits + 10 against the original.
- **Framework items** marked open in section 1.

Appendix A: how the numbers were worked out

Edge share = 2P − 1, where P is the chance that a given unit of the edge holder activates before a given unit of the opponent, averaged over all pairs. It runs from 0 (a fair bag, P = 0.5) to 1 (the edge holder always first). The earlier "mean rank difference" measure grows with fleet size, so it can't be compared across game sizes; edge share can.

Assumptions:

- Shooting phase only. The edge holder activates a unit as soon as one of its dice is drawn and never passes.
- Each capital ship and each escort squadron is one unit.
- Escort squadrons roll Leadership once, and use summed starting hits.
- Leadership is rolled from Appendix 4: the Default column for Imperial ships and the Eldar column for Eldar. Fleet commanders have fixed Ld.
- Bag results are exact calculations. ABFG results are exact over all Leadership rolls, using ABFG's tie-break: initiative, then the unit's Leadership, then the admiral's Leadership (equal in every matchup here: Ld9 against Ld9, or none at 500 points), then attack rating (Corsair Eldar 4, Imperial Navy 2), so remaining ties go to the Eldar.
- 500-point lists have no fleet commander (one is required only above 750 points).

ABFG against the gap. To see what ABFG gives at each gap, one fleet's initiatives were shifted by whole numbers and the edge share recomputed, for all 91 matchups. The mean gap predicts ABFG's edge only loosely, because ABFG depends on the whole spread of initiatives. A fleet of a few low-initiative capital ships against many high-initiative units gives a large ABFG edge at a modest gap. Ties going to the Eldar also give them an edge at a gap of 0.

Gap	0	1	2	3
ABFG edge share (range over 91 matchups)	−0.12 to 0.27	0.10-0.62	0.35-0.84	0.50-0.95

So the gap at which ABFG matches a given dice bonus spreads widely:

Bonus	ABFG-equivalent gap
+1 die per 3 units	−0.2 to 1.6
+1 die per 2 units	0.1-1.9
+1 die per unit	0.4-2.8
+3 dice per 2 units	0.6-3.8
+2 dice per unit	0.8-3.8
+3 dice per unit	1.5-4.5

No gap table can track ABFG closely across list styles. System 3's bands are set for a moderate, predictable edge, and are not an ABFG match.

Removing the die on the choice. If the side that takes the free first activation does not remove one of its dice, it is left with a surplus die that works exactly like a bonus die. At 5 units that raises the choice from 0.20 to 0.33.

Appendix B: test fleets

Imperial lists from the Armageddon Sector list; Eldar lists from the MMS Corsair list, within their fleet limits. Fleet commanders: Fleet Admiral (Ld9) and Shadow Prince (Ld9) at 1000 points and above; none at 500 points. No secondary commanders, refits or upgrades.

Points	Imperial gunline	Imperial carrier/escort
500	Lunar, Gothic, 4 Swords	Dictator, Dauntless, 3 Swords, 2 Cobras (485)
1000	Lunar (Fleet Admiral), Gothic, Dominator, Dauntless, 4 Swords, 4 Cobras (995)	
1500	Retribution (Fleet Admiral), Mars, Dominator, Gothic, Lunar, Tyrant, 2 Cobras (1475)	Emperor (Fleet Admiral), Dictator, Lunar, Gothic, Dauntless, 4 Swords, 4 Firestorms, 3 Cobras (1490)
2000	Emperor (Fleet Admiral), Retribution, Mars, Overlord, Dominator, Gothic, Lunar, Tyrant	Emperor (Fleet Admiral), Mars, 2 Dictators, Lunar, 2 Dauntless, 4 Swords, 4 Firestorms, 6 Cobras (1980)

Points	Eldar gunline	Eldar escort
500	Shadow, Aurora, Solaris (495)	Aurora, 3 Hellebores, 2 Hemlocks (465)
1000	Eclipse (Shadow Prince), Shadow, Aurora, 2 Hellebores, 3 Nightshades (980)	
1500	Void Stalker (Shadow Prince), Eclipse, Shadow, Aurora, Solaris, 2 Hellebores (1425)	Eclipse (Shadow Prince), Shadow, 2 Auroras, 3 Hellebores, 3 Nightshades, 2 Hemlocks, 3 Aconites (1475)
2000	2 Void Stalkers (one with Shadow Prince), Supernova, Eclipse, Shadow, Aurora, 2 Nightshades (1990)	Eclipse (Shadow Prince), 2 Shadows, Aurora, Solaris, 3 + 2 Hellebores, 4 Nightshades, 3 Hemlocks, 3 Aconites (1940)

Escort builds (same ships, three ways)

Points	Imperial: non-escort ships	Imperial escorts	Pairs	3s and 4s	Massed
500	Dauntless	6 Swords, 6 Cobras (500)	6 pairs	4 squadrons of 3	2 squadrons of 6
1000	Lunar (Fleet Admiral), Dauntless	6 Swords, 6 Firestorms, 6 Cobras (965)	9 pairs	6 squadrons of 3	3 squadrons of 6
1500	Emperor (Fleet Admiral), Dictator, Lunar, Gothic, 2 Cobras	6 Swords, 6 Firestorms (1490)	6 pairs (+ the Cobra pair)	4 squadrons of 3 (+ the Cobra pair)	2 squadrons of 6 (+ the Cobra pair)
2000	Emperor (Fleet Admiral), Mars, Dictator, Lunar, Gothic, Dauntless	6 Swords, 6 Firestorms, 6 Cobras (1980)	9 pairs	6 squadrons of 3	3 squadrons of 6

Points	Eldar: non-escort ships	Eldar escorts	Pairs	3s and 4s	Massed
500	Aurora	6 Nightshades (440)	3 pairs	2 squadrons of 3	1 squadron of 6
1000	Eclipse (Shadow Prince), Aurora	6 Nightshades, 4 Hemlocks (950)	5 pairs	3, 3 and 4	squadrons of 6 and 4
1500	Eclipse (Shadow Prince), Shadow, Aurora	6 Nightshades, 4 Hellebores, 4 Hemlocks (1480)	7 pairs	3, 3, 4 and 4	squadrons of 6, 4 and 4
2000	Eclipse (Shadow Prince), 2 Shadows, Aurora	6 Nightshades, 6 Hellebores, 6 Hemlocks (1960)	9 pairs	6 squadrons of 3	3 squadrons of 6

The 500-point Eldar escort build is 440 points; no further escort fits all three builds. MMS gives no escort squadron size limit; squadrons here are kept to XR's 2-6.

Appendix C: all 91 matchups

Gap = Eldar mean − Imperial mean. Positive shares favour the Eldar, negative the Imperials. ABFG uses its tie-break chain. Shooting phase only. Orks are not modelled yet.

Points	Imperial list (units)	Eldar list (units)	Gap	ABFG	System 3 gives	System 3
500	gunline (3)	gunline (3)	+3.0	+0.78	Eldar: choice + 2	+0.60
500	gunline (3)	escort (3)	+4.7	+0.93	Eldar: choice + 2	+0.60
500	gunline (3)	escorts in pairs (4)	+5.2	+0.96	Eldar: choice + 2	+0.50
500	gunline (3)	escorts massed (2)	+2.7	+0.72	Eldar: choice + 1	+0.67
500	gunline (3)	escorts in 3s and 4s (3)	+4.4	+0.91	Eldar: choice + 2	+0.60
500	carrier/escort (4)	gunline (3)	+1.0	+0.32	Eldar: choice	+0.33
500	carrier/escort (4)	escort (3)	+2.7	+0.70	Eldar: choice + 1	+0.50
500	carrier/escort (4)	escorts in pairs (4)	+3.2	+0.77	Eldar: choice + 2	+0.50
500	carrier/escort (4)	escorts massed (2)	+0.7	+0.24	Eldar: choice	+0.50
500	carrier/escort (4)	escorts in 3s and 4s (3)	+2.4	+0.66	Eldar: choice + 1	+0.50
500	escorts in pairs (7)	gunline (3)	-1.1	-0.30	Imperial: choice	-0.14
500	escorts in pairs (7)	escort (3)	+0.6	+0.35	Eldar: choice	+0.33
500	escorts in pairs (7)	escorts in pairs (4)	+1.1	+0.55	Eldar: choice	+0.25
500	escorts in pairs (7)	escorts massed (2)	-1.4	-0.40	Imperial: choice	-0.14
500	escorts in pairs (7)	escorts in 3s and 4s (3)	+0.3	+0.22	No edge holder: plain bag	0.00
500	escorts massed (3)	gunline (3)	+2.3	+0.87	Eldar: choice + 1	+0.50
500	escorts massed (3)	escort (3)	+4.0	+0.98	Eldar: choice + 2	+0.60
500	escorts massed (3)	escorts in pairs (4)	+4.5	+0.99	Eldar: choice + 2	+0.50
500	escorts massed (3)	escorts massed (2)	+2.0	+0.83	Eldar: choice + 1	+0.67
500	escorts massed (3)	escorts in 3s and 4s (3)	+3.7	+0.98	Eldar: choice + 2	+0.60
500	escorts in 3s and 4s (5)	gunline (3)	-0.1	+0.13	No edge holder: plain bag	0.00
500	escorts in 3s and 4s (5)	escort (3)	+1.6	+0.66	Eldar: choice + 1	+0.50
500	escorts in 3s and 4s (5)	escorts in pairs (4)	+2.1	+0.69	Eldar: choice + 2	+0.50
500	escorts in 3s and 4s (5)	escorts massed (2)	-0.4	-0.01	No edge holder: plain bag	0.00
500	escorts in 3s and 4s (5)	escorts in 3s and 4s (3)	+1.3	+0.66	Eldar: choice	+0.33
1000	original (6)	original (5)	+3.0	+0.78	Eldar: choice + 3	+0.50
1000	original (6)	escorts in pairs (7)	+4.3	+0.92	Eldar: choice + 4	+0.45
1000	original (6)	escorts massed (4)	+2.2	+0.73	Eldar: choice + 2	+0.50
1000	original (6)	escorts in 3s and 4s (5)	+3.2	+0.86	Eldar: choice + 3	+0.50
1000	escorts in pairs (11)	original (5)	-0.3	-0.03	No edge holder: plain bag	0.00
1000	escorts in pairs (11)	escorts in pairs (7)	+1.0	+0.45	Eldar: choice	+0.14
1000	escorts in pairs (11)	escorts massed (4)	-1.1	-0.35	Imperial: choice	-0.09
1000	escorts in pairs (11)	escorts in 3s and 4s (5)	-0.1	+0.02	No edge holder: plain bag	0.00
1000	escorts massed (5)	original (5)	+3.0	+0.93	Eldar: choice + 3	+0.50
1000	escorts massed (5)	escorts in pairs (7)	+4.3	+0.99	Eldar: choice + 4	+0.45
1000	escorts massed (5)	escorts massed (4)	+2.2	+0.91	Eldar: choice + 2	+0.50
1000	escorts massed (5)	escorts in 3s and 4s (5)	+3.2	+0.98	Eldar: choice + 3	+0.50
1000	escorts in 3s and 4s (8)	original (5)	+0.7	+0.28	Eldar: choice	+0.20
1000	escorts in 3s and 4s (8)	escorts in pairs (7)	+2.0	+0.59	Eldar: choice + 3	+0.40
1000	escorts in 3s and 4s (8)	escorts massed (4)	-0.1	+0.07	No edge holder: plain bag	0.00
1000	escorts in 3s and 4s (8)	escorts in 3s and 4s (5)	+0.9	+0.45	Eldar: choice	+0.20
1500	gunline (7)	gunline (6)	+3.2	+0.66	Eldar: choice + 3	+0.44
1500	gunline (7)	escort (8)	+4.7	+0.84	Eldar: choice + 4	+0.42
1500	gunline (7)	escorts in pairs (10)	+5.6	+0.89	Eldar: choice + 5	+0.40
1500	gunline (7)	escorts massed (6)	+3.6	+0.76	Eldar: choice + 3	+0.44

1500	gunline (7)	escorts in 3s and 4s (7)	+4.3	+0.81	Eldar: choice + 4	+0.45
1500	carrier/escort (8)	gunline (6)	+2.1	+0.51	Eldar: choice + 2	+0.38
1500	carrier/escort (8)	escort (8)	+3.6	+0.78	Eldar: choice + 4	+0.42
1500	carrier/escort (8)	escorts in pairs (10)	+4.5	+0.84	Eldar: choice + 5	+0.40
1500	carrier/escort (8)	escorts massed (6)	+2.5	+0.64	Eldar: choice + 2	+0.38
1500	carrier/escort (8)	escorts in 3s and 4s (7)	+3.2	+0.75	Eldar: choice + 4	+0.45
1500	escorts in pairs (11)	gunline (6)	+0.1	+0.02	No edge holder: plain bag	0.00
1500	escorts in pairs (11)	escort (8)	+1.6	+0.29	Eldar: choice + 3	+0.36
1500	escorts in pairs (11)	escorts in pairs (10)	+2.5	+0.54	Eldar: choice + 4	+0.36
1500	escorts in pairs (11)	escorts massed (6)	+0.5	-0.04	Eldar: choice	+0.17
1500	escorts in pairs (11)	escorts in 3s and 4s (7)	+1.2	+0.16	Eldar: choice	+0.14
1500	escorts massed (7)	gunline (6)	+2.6	+0.59	Eldar: choice + 2	+0.38
1500	escorts massed (7)	escort (8)	+4.1	+0.82	Eldar: choice + 4	+0.42
1500	escorts massed (7)	escorts in pairs (10)	+5.0	+0.89	Eldar: choice + 5	+0.40
1500	escorts massed (7)	escorts massed (6)	+3.0	+0.73	Eldar: choice + 3	+0.44
1500	escorts massed (7)	escorts in 3s and 4s (7)	+3.7	+0.79	Eldar: choice + 4	+0.45
1500	escorts in 3s and 4s (9)	gunline (6)	+1.1	+0.25	Eldar: choice	+0.17
1500	escorts in 3s and 4s (9)	escort (8)	+2.6	+0.54	Eldar: choice + 3	+0.36
1500	escorts in 3s and 4s (9)	escorts in pairs (10)	+3.5	+0.65	Eldar: choice + 5	+0.40
1500	escorts in 3s and 4s (9)	escorts massed (6)	+1.5	+0.27	Eldar: choice + 2	+0.38
1500	escorts in 3s and 4s (9)	escorts in 3s and 4s (7)	+2.2	+0.46	Eldar: choice + 3	+0.40
2000	gunline (8)	gunline (7)	+3.3	+0.71	Eldar: choice + 4	+0.45
2000	gunline (8)	escort (10)	+5.7	+0.99	Eldar: choice + 5	+0.40
2000	gunline (8)	escorts in pairs (13)	+6.8	+0.99	Eldar: choice + 7	+0.40
2000	gunline (8)	escorts massed (7)	+4.2	+0.97	Eldar: choice + 4	+0.45
2000	gunline (8)	escorts in 3s and 4s (10)	+5.9	+0.99	Eldar: choice + 5	+0.40
2000	carrier/escort (10)	gunline (7)	+1.3	+0.34	Eldar: choice	+0.14
2000	carrier/escort (10)	escort (10)	+3.7	+0.87	Eldar: choice + 5	+0.40
2000	carrier/escort (10)	escorts in pairs (13)	+4.8	+0.92	Eldar: choice + 7	+0.40
2000	carrier/escort (10)	escorts massed (7)	+2.2	+0.70	Eldar: choice + 3	+0.40
2000	carrier/escort (10)	escorts in 3s and 4s (10)	+3.9	+0.87	Eldar: choice + 5	+0.40
2000	escorts in pairs (15)	gunline (7)	-1.1	-0.14	Imperial: choice	-0.07
2000	escorts in pairs (15)	escort (10)	+1.3	+0.26	Eldar: choice	+0.10
2000	escorts in pairs (15)	escorts in pairs (13)	+2.4	+0.55	Eldar: choice + 5	+0.33
2000	escorts in pairs (15)	escorts massed (7)	-0.2	-0.14	No edge holder: plain bag	0.00
2000	escorts in pairs (15)	escorts in 3s and 4s (10)	+1.5	+0.30	Eldar: choice + 4	+0.36
2000	escorts massed (9)	gunline (7)	+1.9	+0.46	Eldar: choice + 3	+0.40
2000	escorts massed (9)	escort (10)	+4.3	+0.96	Eldar: choice + 5	+0.40
2000	escorts massed (9)	escorts in pairs (13)	+5.4	+0.98	Eldar: choice + 7	+0.40
2000	escorts massed (9)	escorts massed (7)	+2.8	+0.89	Eldar: choice + 3	+0.40
2000	escorts massed (9)	escorts in 3s and 4s (10)	+4.5	+0.97	Eldar: choice + 5	+0.40
2000	escorts in 3s and 4s (12)	gunline (7)	+0.0	+0.05	No edge holder: plain bag	0.00
2000	escorts in 3s and 4s (12)	escort (10)	+2.4	+0.57	Eldar: choice + 4	+0.36
2000	escorts in 3s and 4s (12)	escorts in pairs (13)	+3.5	+0.67	Eldar: choice + 7	+0.40
2000	escorts in 3s and 4s (12)	escorts massed (7)	+0.9	+0.18	Eldar: choice	+0.14
2000	escorts in 3s and 4s (12)	escorts in 3s and 4s (10)	+2.6	+0.62	Eldar: choice + 4	+0.36